Roja

COAL Assignment #01 Date:

(a) In real addressing mode, a 20-bit physical address is generated by shifting the 16 bit segment register by 4 bits.

(Method physical address = (segment 10h) + offset

Pair 1: 4A5C : 0007
segment 4A5Ch → shifted : 4A5C0h
offset : 0007h

4A5C0h
0007h
4A5C7h

Pair 2: 4AC0 : 05C7
segment 4A00h → shifted : 4A000h
offset : 05C7h

4A000h
05C7h
4A5C7h

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(b) Logical address (segment : offset) : This is an architectural abstraction consisting of two 16-bit values. It is the address as seen by the programmer, the assembler, and the CPU's execution unit. It allows for memory segmentation, meaning a program's code and data can easily relocate in memory simply by changing the base segment value, without altering the internal offsets.

Physical address : This is the 20-bit hardware signal that the CPU's pin transmits onto physical address bus to select a specific byte in the RAM chip

During assembly, the assembler translates instructions and calculates offsets relative to the start of a segment, but it does not know where the program will physically reside in RAM.

During linking/loading, the linker resolves external references, and the OS loader places the program into available RAM, loading the resulting segment values into the segment registers. However, it still does not calculate a final 20 bit address for each instruction.

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During execution, the actual calculation is done dynamically as the they by the CPU's bus interface unit (BIU) hardware. Right before fetching an instruction or reading/writing a data ~~stored~~ operand, the BIU takes the current value in the relevant segment register (either CS or DS), shifts it left by 4 bits, and adds the offset to generate the 20-bit signal for the address bus.

Q2.

1- AX - B
2- BX - E
3- CX - H
4- DX - F
5- CS - C
6- DS - G
7- SS - A
8- ES - D

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Q3.

(a) little-endian system

the least significant byte (LSB) is stored at the lowest address.

32-bit hex value: 00A34F00h

(b) Big-endian system

The MSB is stored at its lowest address.
32-bit hex value: 004FA300h

(c) If little endian machine sends a multi-byte integer to a big-endian machine byte-by-byte without any conversion, the receiving machine will read the bytes in the reverse order. of what was intended, resulting in a completely incorrect numerical value.

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Q4.

Instruction		Destination Register name	CF	SF
MOV	AX, 05F0H	AX = 05F0h	0	0
ADD	AX, 0A20H	AX = 1010h	0	0
MOV	BX, 0002H	BX = 0002h	0	0
SUB	BX, 0003H	BX = FFFFh	1	1
INC	AX	AX = 1011h	1	0
DEC	BX	BX = FFFEh	1	1

Q5 Bookstore billing

(a) Repeated addition

```
MOV AX, 0
ADD AX, 150
ADD AX, 150
ADD AX, 150
ADD AX, 150
ADD AX, 150
ADD AX, 150
```

(b) JNB AX, 100

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(c) Memory storage (little endian)

The final bill is 650 decimal, which is 028A in hexadecimal.

- 5000h : 8Ah (LSB)
- 5001h : 02h (MSB)

(a) Compute change

```
MOV BX, 1000
SUB DX, AX
```

final register value, BX = 0350h (350 in decimal)

(e) take up unnecessary memory because of repeated add instruction and manually writing it 9 times. Highly tedious and prone to errors. It is solved by MUL instruction.

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Q6. Assembly Declaration

(a) SCORES DB 6 DUP(0)
(b) MARVAL DW 9999
(c) NAME DB "COMSATS", 00h
(d) BUFFER DD 4 DUP

Line	Classification	Translated directly to machine code
1	Directive	NO
2	Directive	NO
3	Directive	NO (allocates memory, but not an executable)
4	Directive	NO (allocates memory)
5	Directive	NO
6	Directive	NO
7	Directive	Yes
8	Directive	Yes
9	Directive	Yes
10	Directive	Yes
11	Directive	NO
12	Directive	NO

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Q8.

(i) 16 chips $\frac{1024}{64} = 16$

4 address lines $\log_2(16) = 4$

(ii) Hardware drivers and Bootloaders

↓

high level lack the ability to directly manipulate CPU control registers.

(iii) Portability. The ability of a program to be compiled and executed on different computer architectures without changing the source code.

Assembly is not portable because it relies on a one-to-one mapping with a specific CPU's instruction set architecture (ISA). The code written on x86 simply does not exist on an ARM processor.

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iv. It is an abstraction layer (like an OS or interpreter) that hides the complex, raw hardware from the programmer. It presents a simplified, logical interface (a virtual machine) making it easier to write programs without managing physical constraints like disk tracks or memory sectors.

v. Even in a single source file a linker is required to calculate and assign the formal relative memory address (offsets) for segments and to attach the necessary executable headers (like the .exe structure) so the operating system knows how to load it.

vi. • Real mode
limits addressing to 1 MB, allows direct and unrestricted hardware access, and has no memory protection (a crash takes down the whole system).
• Protected mode
allows addressing at 4 GB or more, support virtual memory, and enforces hardware-level privileges to prevent user programs from interfering with the OS.

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(vii) It prevents parsing ambiguity. If an identifier began with an digit, the assembler's lexical analyzer would not immediately know whether to treat it as a hexadecimal numeric constant or a variable name. Enforcing a letter first simplifies parsing.

(viii) without overlap : Fetch (2 cycle) + Execute (4 cycles) = 5 cycles total.

with overlap: The BIU fetches the next instruction during the 4 cycles the EU is busy executing. The fetch time is completely hidden (0 extra cycles). The total time remains 4 cycles, significantly increasing throughput.

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